Brick Atlas: From Static Repair to Autonomous Recovery in Brick Assembly

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Figure 1. **Real autonomous construction and a shared repair world.** Top: Claude Sonnet 5.5 builds a car in an archived Arena v2 API run; all six accepted placements and the explicit finish are shown. Decision indices include eight observations and two invalid requests omitted from this image strip but retained in the log. These are model actions, not reference actions; the current v3 scores remain unmeasured. Bottom: eight current-release object examples. The release contains 310 objects and 4,685 static/dynamic questions.

Abstract

Repairing a visible defect and recovering during ongoing assembly place different demands on an agent. The latter requires deciding when to observe, noticing that completed work has changed, and revising a plan while continuing toward the same target. We introduce Brick Atlas, a benchmark linking static repair to autonomous recovery in a shared brick environment. Its current release contains 1,600 static and 3,085 dynamic questions across 310 object configurations, including 57 complete builds and 253 finishing tasks with supplied context. Agents request views, inspect optional instructions, and manipulate one brick per action. Acceptance depends on the final physical arrangement and legal execution, allowing alternative construction orders. We distinguish construction success, actual incident exposure, qualified recovery, and technical failures. All 4,685 published questions pass reference replay; 251 community objects also admit a verified alternative order. These checks establish executable tasks, not model competence or human readability. We provide the evaluation protocol and unmeasured experiment tables for the remaining independent human and learned-agent studies.

1. Introduction

An assembly agent has installed several supports and is preparing the next layer. Between actions, one earlier brick disappears. The intended object has not changed, but the agent's plan may now rely on a structure that no longer exists. Continuing successfully requires more than selecting the next instruction: the agent must acquire useful evidence, update its understanding of the scene, and restore the structure through legal actions.

This distinction motivates our progression from static repair to dynamic recovery. In a static task, a supplied state and one acceptance target define the defect to resolve. In a dynamic task, the agent takes control of an uninterrupted episode: it chooses views, places and removes parts, consults instructions when useful, and decides when to finish. External incidents are changes to this same world. The harness does not pause assembly to ask a separate error-classification question.

Evaluation can otherwise confound several failures. If an agent cannot complete the object without an incident, an intervened failure is weak evidence about recovery. If an incident never occurs, it cannot be counted as missed. If another legal assembly order reaches the target, intermediate

disagreement with a reference should not invalidate it. Finally, a server rejection or a malformed response must not be silently relabeled as a visual recognition error. These considerations determine both the interface and the denominators of our metrics.

Visual ambiguity is equally consequential. A single final image can conceal joints, while an isolated crop from a large model may have no recognizable purpose. We therefore use complete small objects or a supplied complete body with a marked finishing region. The static target is the required stage, even when a later instruction depicts the finished object. Requested current and target views share the same camera and canvas. Candidate previews expose placements without declaring their legality or correctness.

Prior work addresses buildable brick generation [1, 8], manual execution [14], single-step correction [13], and interactive reconstruction [12]. Our focus is the relationship between restoring an explicitly supplied state and maintaining correctness when an agent controls a continuing assembly process. The contribution is a reproducible combination of task construction, active observation, intervention accounting, and repair evaluation; neither interaction nor error correction alone is new.

We contribute (1) a source-traceable release of 310 object configurations and 4,685 checked questions, with historical and rejected records kept separate; (2) an Arena exposing observation and manipulation as explicit tools, with legal alternative orders and endpoint-based scoring; and (3) metrics and executable checks separating repair, damage, exposure, and technical validity. The current manuscript reports dataset and executor evidence. Formal model comparisons, independent human qualification, and independent physical validation remain outstanding.

2. Related Work

Generation and manual execution. BrickGPT generates physically stable and buildable structures from text [8]. BrickNet uses a graph representation of bricks and connections for generative assembly [1]; BrickBench evaluates agentic brick design [2]. MEPNet translates visual assembly manuals into component placements [14]. These tasks establish useful representations and executors, but differ in what specifies success: our target is supplied and remains fixed during an incident. We evaluate restoration and continued completion, rather than the quality of a newly designed object.

Correction and interactive reconstruction. SCANet introduces single-step assembly error correction using manual images and assembled components [13]. Break and Make lets an agent inspect and disassemble an object before rebuilding it with low-level actions [12]. We share the need to connect visual evidence with executable part operations.

Table 1. **Related task contracts.** This comparison describes objectives and interaction, not a ranking of benchmark quality.

Setting	Evidence / goal	Agent response
BrickGPT	Text specification	Generated build
MEPNet	Manual images	Component poses
SCANet	Manual + assembly	Step correction
Break and Make	Interactive object	Inspect, break, rebuild
Brick Atlas static	Current + stage target	Accept or legally repair
Brick Atlas dynamic	Requested views + goal	Observe, build, recover

Our dynamic condition adds unannounced changes during the agent's own continuing trajectory, while the static condition starts directly from a state requiring acceptance or repair. Scores across these contracts are not interchangeable.

Spatial reasoning and task scope. LEGO-Puzzles tests multi-step spatial reasoning [10], and PhyBlock studies progressive physical understanding and planning through block assembly [6]. Spatial457 separates spatial capabilities and analyzes attribute bias [15]. These motivate structured analyses rather than a single pooled score. We additionally separate the amount of assembly, access-related rework, and repeated incidents. Length is a measurable descriptor; we do not equate it with empirically established difficulty.

Tools, embodiment, and recovery. BC-Bench separates brick selection and pose estimation [5]. BrickWright combines an assembly API with feasibility feedback and audits validator discrepancies [9]. WorkBenchMark evaluates perception, planning, and manipulation in Duplo assembly [7]; AssemblyBench tests part motion trajectories with physics-based execution [3]. These motivate explicit interface assistance and independent action checks. AsgardBench studies visually grounded interactive planning with minimal feedback [11], while RoboRecover compares normal execution and recovery from deviations [4]. Our narrower contribution connects static repair and continuing recovery in the same brick world, with active sensing and explicit exposure gates. We do not claim the first interactive or recovery benchmark.

3. Dataset Construction

3.1. Source objects and admitted tasks

The source pool comprises 437 authored community objects from 338 original sets; 99 entries are authored alternate builds. These are acquisition counts, not runnable task

counts. The current executable release admits 263 community configurations and retains 47 earlier designed whole objects. Together they define 310 configurations in 234 source groups. There are 57 complete builds and 253 finishing tasks. Every configuration has one normal dynamic control.

For a complete build, the agent starts with a white studded baseplate and a finite inventory. For a finishing task, the already assembled body remains visible and fixed, while the operable region is indicated. Source-file completeness means preserving all original source instances, colors, and poses across the fixed and operable portions; it does not establish retail set completeness. We retain original files, attribution, scope indices, and hashes. Community sources containing minifigure parts are excluded from this pool.

A model is admitted only after its normal sequence and published repair references execute successfully under the current engine. A failed candidate stays in the audit ledger rather than being counted as another benchmark model. For example, one camper source has an unresolved original color value and remains withheld. Of the retained designed-object candidates, 30 incident conditions are withheld because the current reference does not produce a fully verified effective incident. One additional static bus case is withheld because its defect is invisible in all three default views. These qualification failures are not model failures.

3.2. Question construction and error coverage

Static S0 questions present an already correct stage; S1 questions require repair. Dynamic D0 questions are uninterrupted normal construction, D1 questions schedule one incident, and D2 questions schedule repeated incidents. Table 2 gives the current counts. The community subset contains two to six static questions per object, including a normal control, selected for distinct physical states. Retained earlier objects contribute additional stage and error variants. These variants are useful conditions but not independent source objects.

The question families include missing parts, extra material, wrong color, wrong part, displaced or rotated parts, swaps, repeated or skipped steps, early covering, buried repair, and combinations. We distinguish a physical discrepancy from the work needed to reach it. A wrong-colored piece can be immediately accessible or lie under correct covers. A repeated incident is an episode condition rather than another basic visual defect. Family counts are reported with their actual support; unsupported combinations are not synthesized merely to complete a rectangular table.

Errors are informed by assembly failure patterns, but the generated incidence rates are design choices. They do not estimate how often people make these mistakes. Previously excluded human-error categories remain excluded. Inventory-substitution and articulated-mechanism diagnostics retained in historical banks are not silently counted in the current 310-object release.

3.3. Length, diversity, and source separation

We stratify configurations by active target parts: short (3–7), medium (8–15), and long (16 or more). The release contains 143, 147, and 20 configurations in these strata. A large background model may still have a short finishing task. Occlusion, candidate ambiguity, access depth, and the number of repeated incidents can make two equal-length tasks substantially different; these require separate analyses and human calibration.

We keep authored alternatives from the same original set in one source group. Current development, validation, and test labels separate groups, but every admitted source has been exposed during authoring. Therefore, these labels do not establish untouched-source generalization. A later external-source study must acquire and qualify genuinely unused sources before model selection. Public LDraw files may also have appeared in pretraining.

An exact structural duplicate audit normalizes color, translation, and the 24 axis-aligned rotations; it finds no duplicate groups across the checked source pool and retained designs. This does not certify perceptual or semantic diversity. Family-level imbalance remains substantial, including many road vehicles and buildings. We report object- and source-level summaries so repeated questions do not make this imbalance invisible.

4. Autonomous Assembly Arena

4.1. State, goal, and legal actions

Let s_t denote the actual arrangement and G the acceptance target. Operable parts have anonymous inventory handles and registered placement handles. A placement signature $\sigma(s)$ is a multiset of part type, color, and world pose. It ignores interchangeable actor identities. For the released exact-assembly tasks, valid completion requires an explicit finish, $\sigma(s_T) = \sigma(G)$, and the task's final connection constraints. Fixed scenery is excluded from placement metrics.

The agent acts on its public history h_t, not the private scene state:

$$a_t = \pi(h_t), \quad \widetilde{s}_{t+1} = T(s_t, a_t), \quad s_{t+1} = J_t(\widetilde{s}_{t+1}). \quad (1)$$

Here J_t is identity unless an admissible incident is applied after a physical operation. An observation leaves the arrangement unchanged and returns the requested rendering; manipulation returns feedback. Thus h_t can contain an outdated view of the present state. The scheduler reads the actual scene, while the acting model receives no incident label. This partial observability distinguishes a continuing episode from a list of independently answered snapshots.

Table 2. **Current runnable dataset.** S0/S1 are correct-stage acceptance/static repair; D0/D1/D2 are normal construction/one incident/repeated incidents. Source acquisitions and historical records do not enter these counts.

Scope	Objects	S0	S1	D0	D1	D2	Questions
Complete build	57	41	141	57	1,466	28	1,733
Finishing task	253	253	1,165	253	1,165	116	2,952
Total	310	294	1,306	310	2,631	144	4,685

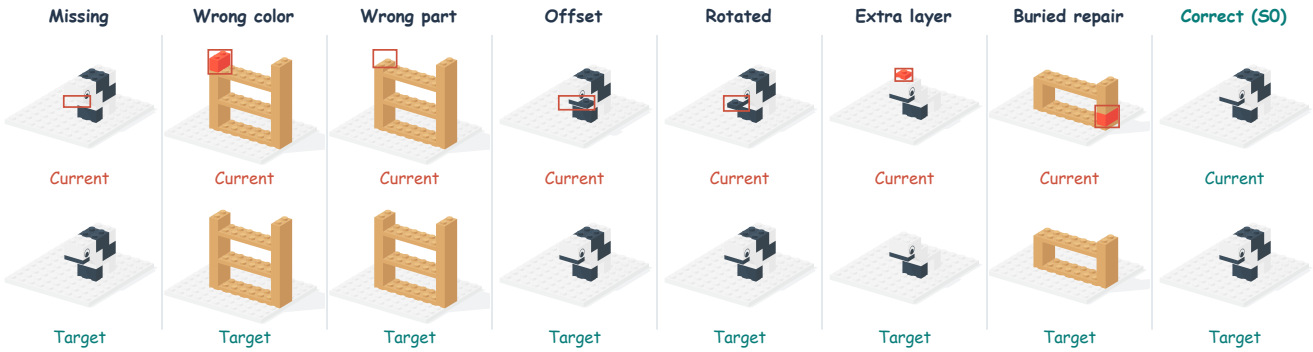


Figure 2. **Eight current static examples.** Each column shows the supplied state above its unique target; matched framing preserves scale. Coral boxes locate changed image regions for readers, not for the model. The correct control requires no repair. Buried repair includes access work; its reference is feasible, not asserted globally shortest. All case IDs and exact geometry states are recorded in the figure manifest.

Each attach or detach moves one physical part. The engine checks compatible connections, overlap, and supported straight insertion or extraction paths. A rejected action leaves the scene unchanged. Reference action sequences demonstrate feasibility and support effort comparisons, but do not define the only correct order. A legal off-target placement can be undone later; it is not automatically an episode failure. Likewise, temporarily removing a correct cover is permissible when it enables repair.

These checks define a constrained assembly world. Fixtures may support a subassembly; gravity, friction, clutch force, gripper reach, and general articulated motion are not simulated. Some background geometry uses conservative obstacles. Thus reference replay establishes engine-relative executability, not an independent proof that a human or robot can perform every motion.

4.2. Requested observation and concrete input/output

The dynamic protocol is `brick-atlas-tool-arena/3`. Each reply contains exactly one `tool` and its `args`. The tools are `observe`, `inspect`, `attach`, `detach`, `recall`, and `finish`. The first request specifies the task, budget, neutral rules, tool schemas, and available anonymous handles. It does not include a batch of scene images.

For example, the model can request angled current and target images while rotating the camera by 90 degrees. Both images are 800 by 600 pixels with matched framing. It can instead request a top view, focus on the operable region or the whole object, or request an inventory page. Inventory pages contain at most twelve items and are accompanied by current installed/available status. Parts and placements are finite; no object can be installed twice by reusing its handle.

The model can inspect a part's hypothetical placement cards or a manual step. Candidate cards depict possible poses without marking which are legal or correct. Instructions give a feasible construction sequence, not mandatory intermediate targets. A recalled image retains its original timestamp and is not a fresh observation. A successful placement provides action feedback only; the model must decide whether another observation is useful.

Static questions use the existing JSON-lines adapter: three matched current/target views and an inventory contact sheet at each decision, then one action object with a `type` field per decision. The stage acceptance target is authoritative. Future manual pages, if requested, are clearly references. This interface difference is explicit: static-to-dynamic comparisons measure the combined demand of autonomy and recovery, not a perfectly isolated intervention effect. Matched-sensor ablations are reserved separately.

4.3. Incidents in the continuing world

An incident is applied after a physical operation when the configured conditions hold in the actual state. The agent's history and remaining budget persist. The scene can lose a

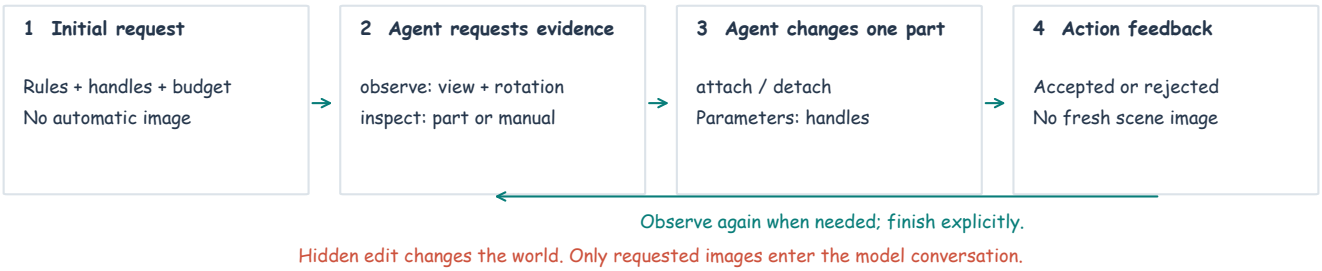


Figure 3. **The model controls sensing and manipulation.** The initial request contains rules, tools, anonymous handles, and budgets, with no automatic image. An observation request returns the requested views; a physical action returns feedback without fresh images. Spectator images and hidden event metadata belong to the review log, not to the model's input.

brick, gain material, or undergo another supported perturbation without a separate notification. Ordinary feedback describes the requested action, not the hidden edit. The next observation reveals the resulting state.

Scheduling follows the agent's actual installed parts and spacing constraints rather than requiring an exact reference prefix. If no admissible effective change is available, the event is deferred or remains untriggered. The evaluator records planned, applied, and verified effective events separately. Repeated-event episodes retain a window for each event, ending just before the next external change. An alternative legal route can therefore encounter an incident at a different point without being penalized for its order.

4.4. Logs and information boundaries

The Arena stores each request, response, requested image hash, action feedback, and terminal result. The spectator page additionally shows multiple scene views and environment interventions. This richer review stream must never be confused with what the model received. Private certificates, reference actions, family names, and hidden event schedules are absent from public requests. Randomized handles are action identifiers, not labels encoding the answer.

A runnable dataset entry points to its model, source group, scope, task type, input policy, and private evaluator files. A public review page may reveal GT after selection; it is not the acting-model endpoint. Release checks compare current question IDs, review GT IDs, Arena IDs, file hashes, and expected image assets. This separation supports reproducible review without rewarding access to private state.

5. Evaluation

5.1. Static correction and preservation

Let $I = \sigma(s_0)$, $F = \sigma(s_T)$, and $G = \sigma(s^*)$ be multisets over operable placements. Define $TP = |F \cap G|$, $FP = |F - G|$, and $FN = |G - F|$. Static repair reports exact

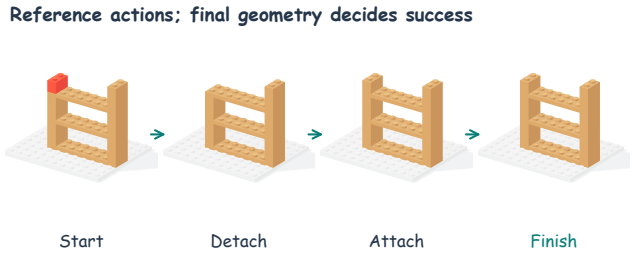


Figure 4. **A feasible repair is a trajectory, not a unique GT order.** This current reference removes the wrong part and restores the required stage. Intermediate progress and final acceptance are scored separately. Correct surrounding work may remain intact. Labels identify executed actions; no learned-agent result is implied.

success SR and placement

$$F_1 = \frac{2TP}{2TP + FP + FN}, \qquad ER = 1 - \frac{FP + FN}{|I - G| + |G - I|}. \tag{2}$$

The residual counts placement discrepancies, not a shortest legal edit path. Wrong color can contribute one excess and one missing placement. Error reduction can be negative if the agent worsens the state. Undefined denominators remain N/A; two empty multisets have $F_1 = 1$.

Let $B = I \cap G$ be initially correct content. Final damage is $|B - F|$, normalized by $|B|$ when nonempty. Temporarily removing and restoring a correct cover incurs effort but no final damage. On S0, false repair FR records any accepted physical move, even if later reversed. Correct acceptance and FR are both reported. The 294 S0 and 1,306 S1 questions are never pooled into a single repair rate.

5.2. Construction-qualified dynamic recovery

D0 establishes normal build completion BCR. Raw intervention completion ICR includes valid assigned runs even

when an event is not reached; exposure is reported alongside it. For each D1 or D2 run, the primary recovery gate requires a successful normal run matched on task, model, input, and budget, a valid terminal status, all scheduled incidents applied, and verified effective changes. Let E be this eligible set. We report

$$\mathrm{NQR} = \frac{1}{|E|} \sum_{i \in E} \mathbf{1}[\text{valid final completion}_i]. \tag{3}$$

NQR corresponds to the implemented `normalQualifiedRecovery` field. Its normal and incident runs are independent continuations. It is a construction-qualified descriptive statistic, not a causal estimate of the incident's effect. Its denominator, unmatched controls, failed controls, untriggered events, partial exposure, and technical exclusions are all reported.

A stricter causal comparison requires actually branching the same saved environment, conversation, observations, handles, and budget. Only verified shared-prefix pairs permit conditional paired recovery or a paired completion difference. Nominally matching seeds does not establish shared history. This paired study is a reserved experiment, not evidence supplied by the current six scripted Arena runs.

5.3. Event recovery and recurrence

For an incident with before/after signatures B and A, define lost correct placements $L = (B \cap G) - A$ and newly introduced excess $X = (A - G) - (B - G)$. A recovery window checks restoration of the lost content and clearance of the incremental excess while allowing additional correct construction. Requiring equality to the pre-event state would incorrectly reject forward progress.

Report first recovery in decisions, recovery at the window end, and retention over at least two later decisions. Observations and invalid attempts count toward decisions. A short window has undefined retention; recurrence is recorded as loss of recovery. The next external incident starts a new window and must not be attributed to the agent undoing an earlier repair. A pure access obstruction without new placement discrepancies uses completion evidence rather than an automatic zero repair latency.

For D2, all-event completion additionally requires every scheduled effective incident and valid final success. Recovery of the first event alone is not successful recovery of the episode. Independent detection or localization accuracy remains unmeasured: a failed episode does not prove that the model failed to see the defect.

5.4. Validity, cost, and aggregation

A wrong explicit finish or model budget exhaustion is an ability failure. Final service failures, harness errors, and external stops are censored from ability rates and reported in the planned-attempt ledger. Parsing errors, schema or handle errors, physical rejection, and resource rejection have separate counters. A successful retry can retain a valid result, but all billed attempts remain in cost.

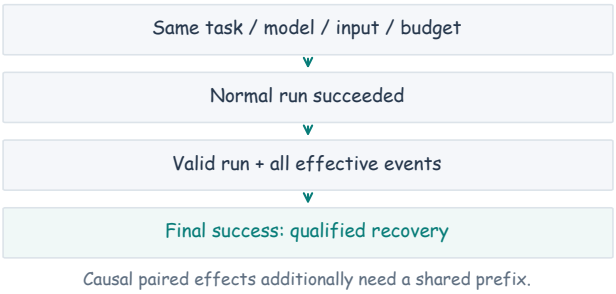


Figure 5. **Repair eligibility is explicit.** Each excluded category retains its own count. Independent normal-qualified recovery and strict shared-prefix paired effects have different evidence requirements. Empty eligible sets produce N/A.

Report task-micro, object-macro, and source-macro results, alongside valid coverage and planned attempts. Variants from a shared source stay together in uncertainty estimation. Costs include requested views, inspected pages, recalls, text and image tokens, retries, and local inference resources where relevant. Moves relative to a feasible reference may be less than one; this is not an optimality score. A single run per model and condition, as planned here, does not establish stochastic repeatability.

6. Release Analysis and Experimental Design

6.1. What the current checks establish

All 4,685 published questions pass reference execution. The source checks preserve every original instance for the 263 admitted community models. A bounded adjacent-swap search verifies a second legal order for 251 of them; absence of a found swap does not prove uniqueness for the remaining twelve. The image audit checks 17,618 assets, current/target canvas agreement, exact normal-control agreement, and nonempty changes for admitted community static errors.

Six scripted Arena episodes cover normal and incident runs on a complete cow, a truck finishing task, and a longer building completion. They exercise requested observation, tool execution, image provenance, qualified recovery, and final scoring. Their success is a harness regression result, not an API-model baseline. The audit does not independently validate visual comprehensibility or mechanical feasibility. Candidate failures, including the 31 withheld legacy conditions, remain visible outside the published denominator.

6.2. Qualitative Arena trajectories

Figure 6 compares two recorded API episodes with feasible reference continuations. These are deliberately selected development cases from Arena v2: one successful recovery and one wrong final submission. Both models previously completed their respective normal builds. Figure 1 shows every accepted car placement; the supplement adds the complete chair construction strip and all decisions. Normal success can include rejected intermediate actions and does not guarantee subsequent recovery. We re-execute every logged action and reproduce every state transition, rejection, and incident; no new API calls are used. Their older handles and input contract prevent treating them as current v3 benchmark scores.

Successful recovery with wasted decisions. In the car episode, Claude Sonnet 5.5 experiences a removal after decision 14, requests a new view at 15, makes an invalid request at 16 and a colliding placement at 17, then observes again. Decision 19 restores the missing target placement; observation and explicit finish follow. Its 21 decisions include nine observations, three format failures, and one physical rejection. Exact final success therefore coexists with inefficient intermediate behavior. A GT continuation from the same faulty state restores one part before finishing.

Activity without correct recovery. For the chair, GPT-6.1 Sol continues after a removal at decision 7, including a requested 90-degree view at 8. It later encounters blocked extraction, removes several installed parts, and rebuilds with a wrong-colored part. At its explicit finish on decision 30, five of seven target placements are correct, with one excess and two missing placements ($F_1 = 10/13$). The GT continuation repairs the removed part and completes the remaining target through legal placements. This trace supports a wrong-final-state diagnosis; it does not establish whether the model internally detected or understood the incident.

6.3. Observed failures and distinct repair demands

Figure 7 summarizes all latest archived pilot conditions, rather than only the two illustrated incidents. Nine valid normal runs yield three successes and six budget terminations; three service failures have no ability outcome. Three incident runs yield one correct and two wrong finishes. The complete audit retains all nineteen attempts, including seven service interruptions. These small, uneven development samples do not establish a model ranking.

Identity and state comparison. Missing/excess content requires counting and correspondence; wrong color or part requires attribute and instance recognition. The chair's final wrong color is an observable identity mismatch, but does not alone distinguish misperception from choosing the wrong handle. Separate description and localization probes test these possibilities without feeding their answers into the primary episode.

Spatial execution and access planning. Offset/rotation demands pose grounding and connection reasoning. A rejected connection or collision identifies an invalid proposed action, not why it was proposed. Buried repair and early covers additionally require ordering access operations. A blocked extraction tests a motion path; wrong final placement tests the endpoint. Supplied-plan execution and legal-pose probes separate these demands under explicitly changed input contracts.

Change tracking and final verification. Swaps and compound edits require reconciling several bindings; repeated incidents add persistent monitoring and retention. Continuing after a removal can reflect stale evidence, a missed discrepancy, or an intentional delayed repair. Only requested evidence and actual repairs are observable. A wrong explicit finish directly exposes an unmet stopping criterion. E32 therefore separates description, localization, pose selection, access order, execution, updating, and completion judgment; assistance effects are diagnostic, not proof of a unique internal cause.

6.4. Questions for the model study

The main study asks whether static correction predicts recovery after successful normal assembly, and how this relationship changes with observation autonomy, access work, length, and repeated incidents. All models receive the same task IDs, budgets, target definitions, and tool permissions within a condition. One run per condition is planned. Account accessibility and exact served revisions must be recorded before results are collected.

The inherited candidate roster comprises GPT-6.1 Sol, Claude Sonnet 5.5, Gemini 3.8 Flash, Claude Haiku 5.5, Qwen3.5-9B, Qwen3.5-27B, InternVL3.5-8B, Qwen2.5-VL-7B, Qwen2.5-VL-32B, and Gemma 3-12B. These are candidate labels, not claims of current availability, completed experiments, or frontier coverage. The supplement retains thirty-two unmeasured result sheets, with identifiers and configuration fields to resolve before execution.

6.5. Three input forms and independent qualification

We specify T (text), I (images), and TI (both) for S0/S1 and D0/D1/D2 (Fig. 8). They change scene evidence, not the goal, finite inventory, action domain, or scorer. Neutral instructions, identifiers, tool schemas, and JSON responses

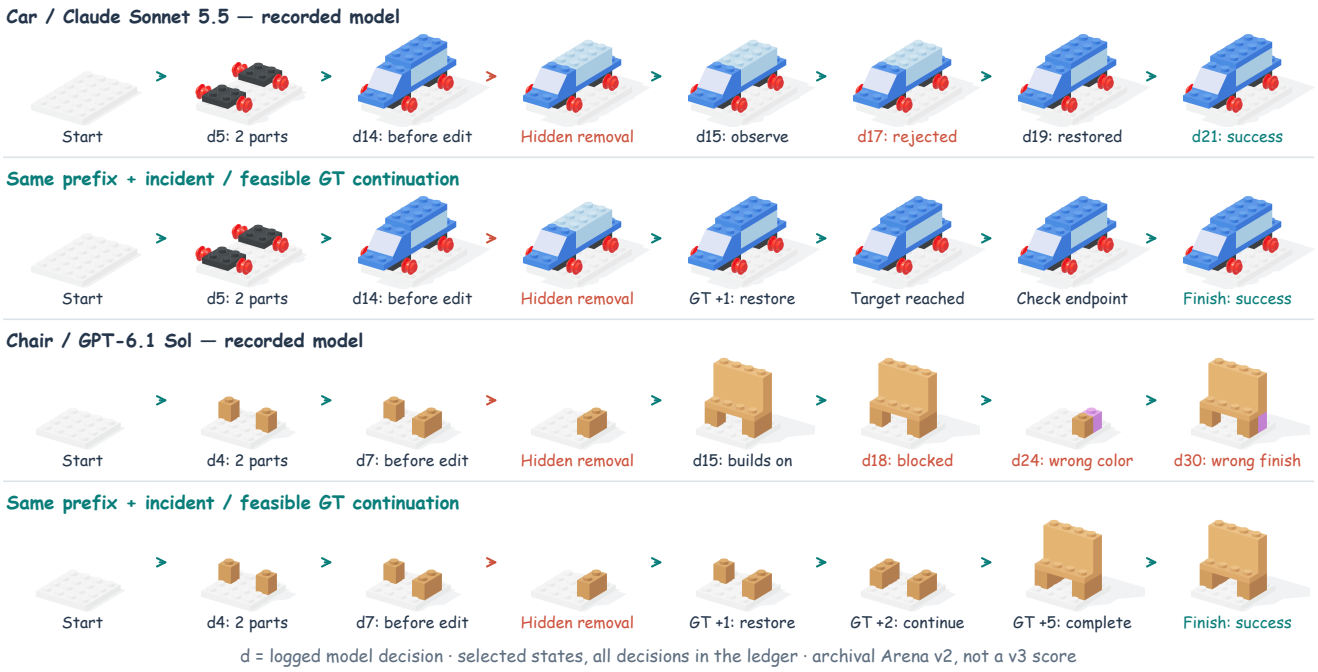


Figure 6. **Actual model behavior versus a feasible GT continuation.** Each row spans an episode using eight selected states; *d* denotes the recorded model decision, including observations and invalid requests. Within each pair, the prefix and hidden removal are identical. The GT row then uses private state to restore the affected part and complete legally; it is neither a learned run nor a shortest-path claim. In the car, the model eventually restores the missing part and finishes. In the chair, it builds further, encounters blocked access, dismantles parts, and finishes with wrong geometry. State images are high-resolution exports with a shared camera, not substituted model inputs. Full decisions and actual requested RGB are provided separately. These archived v2 examples illustrate behavior, not current-release performance or an estimated causal effect.

Observed behavior / all valid latest pilot runs / archived Arena v2

	Format	Handle	No connection	Collision	Blocked path	Wrong finish	Budget limit
Gemini 2.5 Flash / D0 (n=3)	0	3	17	23	0	0	3
GPT-6.1 Sol / D0 (n=3)	0	0	16	6	4	0	2
GPT-6.1 Sol / D1 (n=1)	0	0	1	3	1	1	0
Claude Sonnet 5.5 / D0 (n=3)	9	0	13	0	0	0	1
Claude Sonnet 5.5 / D1 (n=2)	6	0	0	1	1	1	0

First five columns: action counts. Last two: episode counts. Different columns have different denominators.
3 Qwen service failures excluded from ability counts; all 19 attempts (7 service failures) remain in the audit.

Figure 7. **Model behavior, separate from injected error labels.** All twelve valid latest pilot runs, grouped by model and task. Action counts occupy the first five columns; terminal episode counts occupy the last two. D0 and D1 have different exposure and sample sizes. Unknown pose handles are interface errors, not geometric collision evidence. The supplement lists all fifteen latest conditions, including three service failures. These are archived v2 diagnostics, not formal v3 results.

remain common; image-only does not prohibit textual API syntax. T exposes part type, color, pose, and stock in a neutral ledger. I supplies matched scene views and visual inventory/candidate cards. TI supplies both at the same timestamp. Manual access is equal and uses the assigned representation. No channel reveals defect labels, incident times, target-difference lists, or repair plans.

Table 3. **Primary model results (reserved; unmeasured).** Candidate names are inherited planning labels; exact serving versions remain to be verified. Static SR uses S1, FR uses S0, BCR uses D0, and NQR reports fully exposed runs with successful matched normal controls. Each populated rate must include its denominator.

Model	S1 SR	S0 FR	D0 BCR	D1 NQR	D2 NQR	Valid N
GPT-6.1 Sol	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—

Table 4. **Planned analyses and required evidence.** All model outcomes remain unmeasured. The supplementary result sheets separate implemented metrics from protocols requiring additional integration.

Research question	Comparison and outputs	Required control
Does static skill transfer to recovery?	S1 SR/F1, D0 BCR, D1 NQR	Same object; normal success and exposure ledger
Does observation autonomy matter?	Active vs scheduled views; queries and cost	Matched target, resolution, budget, image content
Does access or length increase effort?	Family and length strata; damage and moves	Object/source macro; feasible references only
Does repair remain stable?	D1 vs D2; per-event recovery and retention	Stop each window before the next external edit
What does the representation contribute?	T, I, TI in static and dynamic tasks	Native-input vs matched-fact studies; adapters pending
Can humans read and operate the tasks?	Blind operation plus separate GT review	Independent participants; no answer exposure

For dynamic tasks, `observe` returns the chosen representation only on request; actions still return feedback, and recall retains its old timestamp. Static tasks keep their supplied-evidence schedule. E11 measures static modalities; E31 measures D0 construction and D1/D2 recovery, with normal qualification within each modality. E29 separately crosses modality with fresh/continuing history under matched sensing.

Two claims must stay separate. Native text with exact poses can reveal hidden structure unavailable in RGB, so its gains measure state assistance as well as representation. A matched-fact control instead serializes or rasterizes the same ledger, including stock and occluded facts, in T/I/TI. It tests symbolic encoding and OCR, not ordinary 3D vision. The current interface includes textual stock status alongside RGB and is not a strict I implementation; status must move into visual cards for that arm. These adapters are specified, not yet evaluated.

Similarly, candidate handles and free coordinates pose different localization problems. A coordinate adapter requires a matched action domain and explicit discretization or tolerances before comparison. Independent snapshot detection and supplied-checkpoint recovery require separately admitted tasks. System-supplied progress must not be represented as progress built by the evaluated model.

Human review has two roles. A simple GT review asks whether the target is legible, the defect is visible in available views, and the proposed repair is reasonable. A blind operation study gives only model-visible evidence and tools and measures actual performance. Agreement with a displayed solution is useful audit evidence but cannot substitute for blind recovery. Independent mechanical review must additionally examine occluded connections, extraction paths, and the consequences of removing supporting parts.

6.6. Diagnostic baselines and metric validity

Finish-only, manual-only, and periodic-checker policies anchor the learned-agent study. Privileged witnesses verify

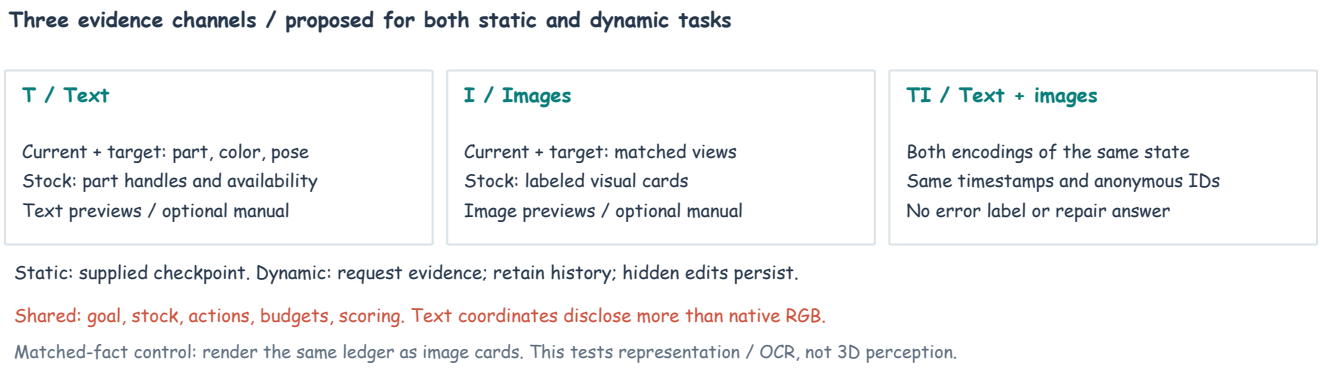


Figure 8. **Proposed modality integration.** Static checkpoints receive the assigned evidence at each decision; dynamic agents request it explicitly, with no automatic state refresh. Exact text and native RGB do not reveal equal information. A separate matched-fact control renders the identical ledger as image cards, testing representation/OCR rather than natural 3D perception. Adapters and formal measurements remain pending.

feasibility and remain separate. Assistance can disclose incident timing, the affected part, symbolic state, or a feasible plan; improvements narrow hypotheses but do not uniquely diagnose internal reasoning. Independent snapshot probes complement these interventions.

Scorer tests should preserve acceptance under independent action swaps and interchangeable handles, charge reversible access work as effort rather than final damage, and reject incorrect endpoints. Independent reviewers adjudicate both accepted and rejected motions. Reference replay alone can share the generator's mistakes. The supplement specifies these checks and their unmeasured tables.

6.7. Process-level measures and controlled transfer

Beyond final success, planned measures include first fresh observation after an incident, recovery within four/eight decisions, retained repair, trajectory F1, and wrong explicit finishes. Observation delay is not detection time; recalled images are stale. Terminal unrepaired states count as failures, while a new event before the scoring horizon is censored and reported. Progress and effort accompany final success: necessary access work can temporarily worsen the state. Full definitions and empty sheets appear in the supplement.

The transfer study crosses fresh versus continuing agents with active versus scheduled sensing at the same saved faulty state. Match the target, inventory, action domain, evidence, and remaining budget; retain prior history only for continuing agents. An equal-total-budget sensitivity includes reaching the checkpoint. Apply this design within each modality and report common-qualified objects alongside full eligibility, rather than subtracting pooled scores from different tasks.

6.8. Limitations and submission status

The long stratum contains only twenty objects; 253 of 310 configurations are finishing tasks. Error support is uneven and all sources were exposed during authoring. Synthetic incidents do not estimate natural human error frequencies. Geometry checks share the implementation used for generation, so independent validation remains necessary. Model interfaces simplify pose selection through finite candidate handles, and the engine omits force dynamics and unrestricted articulation.

The release provides executable tasks, traceable assets, and scoring contracts. A competitive empirical benchmark paper additionally needs current-interface model results, independent human readability and operation evidence, and external checks of the physical rules. The unmeasured tables are a concrete study plan, not evidence that these requirements have been met. Historical pilot traces illustrate behavior in Sec. 6.2 and are not transferred to the present interface.

Data provenance and responsible release. Source attribution and license records accompany the release; redistribution needs source-level rights checks. Public-source pretraining overlap remains possible. Human studies require consent, documented answer exposure, and anonymized judgments. The supplement details these obligations.

7. Conclusion

Brick Atlas connects static repair to autonomous recovery through 310 objects and 4,685 executable questions. Agents control sensing and legal manipulation; acceptance permits alternative orders. Real pilot traces motivate separate studies of recognition, action, access, updating, and stopping. Current-interface model results and independent human and physical validation remain necessary.

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